

Human Development Index (HDI) Predictor

Project Documentation & Technical Report

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1. Abstract & Problem Statement

The Human Development Index (HDI) is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable, and having a decent standard of living. Traditional reporting methods by global bodies are often retrospective, lagging by a year or more, and do not readily offer interactive scenario testing for policy changes.

The **HDI Predictor** addresses this gap by creating an interactive web application powered by an underlying machine learning engine. Using socio-economic indicator inputs, the model estimates the region's overall HDI score dynamically. This allows policy analysts, researchers, and students to evaluate how shifts in specific health, education, or economic metrics might influence absolute human development indices.

2. Features

- **Predictive Engine:** Utilizes a specialized Linear Regression model trained on global human development indicator data to instantly calculate expected HDI scores.
- **Exploratory Data Analysis (EDA):** Built-in visual metrics providing detailed insights into historical distribution curves (such as global HDI distributions) and feature correlations.
- **Interactive Frontend:** A clean and accessible Flask web UI allowing structured form inputs and immediate interactive feedback for predictive simulation.

3. Technology Stack

Python

Flask

Scikit-Learn

Pandas

NumPy

Matplotlib

Seaborn

Pickle

4. Project Directory Structure

```
HDI_Predictor/
├── dataset/
│   └── archive.zip
├── eda/
│   └── graphs.py
├── model/
│   └── model.pkl
├── src/
│   ├── app.py
│   └── train_model.py
├── templates/
│   └── index.html
├── static/
├── screenshots/
│   ├── home_page.png
│   ├── prediction.png
│   └── graphs.png
├── README.md
├── requirements.txt
└── Project_Report.pdf
```

5. Implementation Status & Future Scope

The application is structured around a modular architectural paradigm separating data analysis (eda/), machine learning training operations (src/train_model.py), and the deployment application routing microframework (src/app.py). Initial data passes confirm consistent training distributions across global records.

Future Enhancements: Future scopes include upgrading to multi-variate non-linear regression algorithms (e.g., Random Forest Regressor) to optimize error variances, and integration with live World Bank API endpoints for automatic, real-time yearly baseline data refactoring.